

Econ 101 Fall 2024

Midterm 1

Solutions

MCQ: Permutation Table & Answer Key

Question permutation table:

Question	Version	I	II	III	IV
1	→	6	1	6	5
2	→	3	2	4	3
3	→	1	3	5	4
4	→	2	4	1	6
5	→	4	5	2	1
6*	→	5	6	3	2

*: Question follows previous

Question	Version	I	II	III	IV
1		(d)	(b)	(c)	(c)
2		(a)	(d)	(b)	(a)
3		(b)	(a)	(d)	(b)
4		(a)	(a)	(d)	(c)
5		(a)	(b)	(c)	(b)
6		(b)	(b)	(c)	(a)

MCQ: brief explanations

Medina told Akmaral that since the cinema ticket price increased from 1,450 to 2,900 tenge, she now goes only half as often as she used to. With this information Akmaral knows that:

- (a) Medina is not a true fan of cinema.
- (b) The price elasticity of demand of Medina, in absolute value, is 2.
- (c) The price elasticity of demand of Medina, in absolute value, is $\frac{1}{2}$.
- (d) The price elasticity of demand of Medina cannot be calculated with the data we have.

Correct answers: (c)

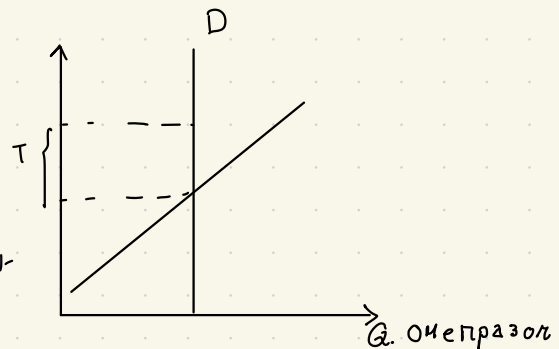
$$\Delta\% P = 1 \Rightarrow \epsilon_D^P = -\frac{1}{2}$$

$$\Delta\% Q = -\frac{1}{2}$$

An NU Economics professor developed gastritis after years of hearing his students say, "Economics is not applicable to real life." His doctor prescribed Omeprazole (Омепразол). He noticed that due to the tax imposed by the government, Omeprazole has become more expensive. Nevertheless, he continues to buy it. We can infer that:

- (a) The professor has an inelastic demand for Omeprazole.
- (b) The professor has an elastic demand for Omeprazole. ✗
- (c) Omeprazole, by not having many substitutes, has an inelastic supply curve.
- (d) Omeprazole, by not having many substitutes, has an elastic supply curve.

Correct answers: (a)



Only (d) deals with opportunity costs.

If an individual lives longer than expected, having spent their savings, and thus requiring financial assistance from their children, then the opportunity cost of the resources they receive from their children is _____. If the individual dies earlier than expected, leaving an inheritance to their children, the corresponding opportunity cost of the resources inherited by their children is _____.

- (a) Not having died when expected. The greater consumption that the children will now be able to enjoy.
- (b) The excess of their own consumption that led them to financially depend on their children. The poorly used savings.
- (c) The interest that they stopped earning because their savings were depleted prematurely. The cost of the funeral.
- (d) At least equal to the unrealized consumption by their children. At least equal to the consumption they did not realize.

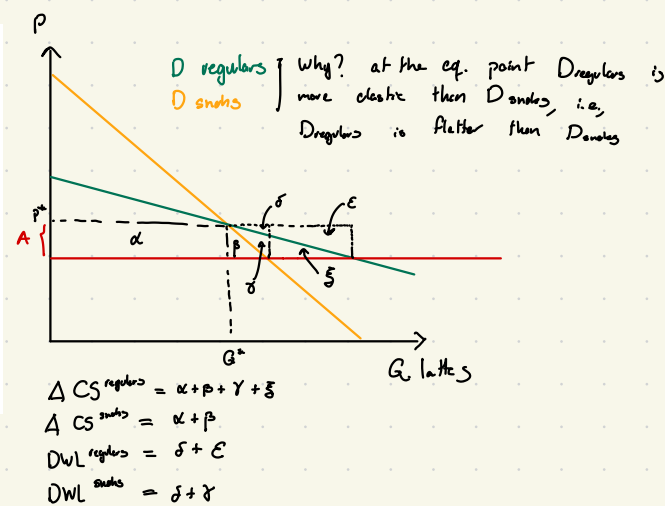
Correct answers: (d)

4. In the town of NUTopia, the townspeople like to consume lattes. Suppose there are two types of consumers: regular coffee drinkers (regulars R), and coffee snobs (snobs S). Suppose that the market equilibrium happens at the same price and quantity for both the regular and snob's market. at the *market equilibrium*, the price elasticity of demand for regular drinkers is $\epsilon_{D,R}^p = -2$ and the price elasticity of demand for snobs is $\epsilon_{D,S}^p = -1.2$. If the supply of lattes in NUTopia is perfectly elastic and if the government of NUTopia implements a subsidy of \$4 in this market we can be sure that: The increase in the consumer surplus for coffee snobs will be _____ than the increase in the consumer surplus for regulars. The dead-weight loss will be greater in the market of _____

- (a) lower than; regulars.
(b) greater than; regulars.
(c) lower than; snobs.
(d) greater than; snobs.

Correct answers: (a)

Answer explanations: (All answers are correct.)



Read carefully the following text and answer questions 5–6.

In Seville, Figaro, the versatile barber, and Dr. Bartolo, the wealthy but conniving guardian of Rosina, are both involved in two key activities: cutting hair and writing love letters. Figaro is excellent at cutting hair and is reasonably good at writing persuasive love letters. Dr. Bartolo, on the other hand, is not great at either task, but he is slightly better at writing letters than cutting hair. In one day:

- Figaro can cut 6 heads of hair (H) or write 4 love letters (L).
- Dr. Bartolo can cut 2 heads of hair (H) or write 2 love letters (L).

In the questions below, put H in the horizontal axis and L in the vertical axis.

5. The production possibilities frontier of Figaro is represented by the equation _____, the production possibilities frontier of Dr. Bartolo is represented by the equation _____. For Figaro the point $(H, L) = (3, 2)$ is _____, for Dr. Bartolo the point $(H, L) = (1, 1)$ is _____.

- (a) $H = 6 - \frac{2}{3}L$; $H = 2 - L$; not feasible; feasible and efficient.
 (b) $H = 6 - \frac{3}{2}L$; $H = 2 - L$; feasible and efficient; feasible and inefficient
 (c) $L = -\frac{3}{2}H + 4$; $L = 2 - H$; not feasible; feasible and efficient.
 (d) $L = -\frac{3}{2}H + 4$; $L = 2 - H$; feasible and efficient; feasible and efficient.

Correct answers: (d)

Answer explanations: (All answers are correct.)

5. Does anyone of them has a comparative advantage? _____. Does any one of them has an absolute advantage? _____. If each of them specializes in the production of the good/service in which they have a comparative advantage, and they decide to trade between themselves, then Figaro would specialize in _____ and Dr. Bartolo in _____, and the possible terms of trade (opportunity cost of H in terms of L when trading) would be _____

- (a) Yes; yes; cutting hair; love letters, $\frac{5}{6}$.
 (b) Yes; yes; love letters; cutting hair, $\frac{6}{5}$.
 (c) There are no comparative advantages.
 (d) Yes; no; cutting hair; love letters; 1.

Correct answers: (a)

C.O. Figaro $(H) = \frac{2}{3} L \rightarrow PPF: L = 4 - \frac{2}{3}H$
 C.O. Bartolo $(H) = 1 \rightarrow PPF: L = 2 - H$

$(3, 2)$ is feasible and on the PPF of Figaro
 $(1, 1)$ is feasible and on the PPF of Bartolo

- (i) Yes: Figaro in H and Bartolo in L
 (ii) Yes: Figaro has an absolute advantage in both H and L (draw the PPF)
 (iii) any number between $[\frac{2}{3}, 1]$ works; here $\frac{5}{6}$ is the choice.

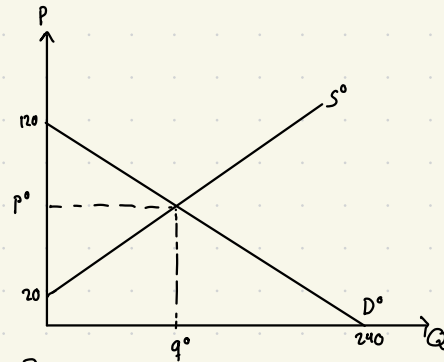
Long Problems

The whimsical land of Ekonomika

Note: Real Economists do not need to be told to graph, they do it intuitively ☺

$$1. \quad Q_d^0(p^0) = Q_s^0(p^0) \Rightarrow p^0 = \frac{160}{3} \approx 53.33$$

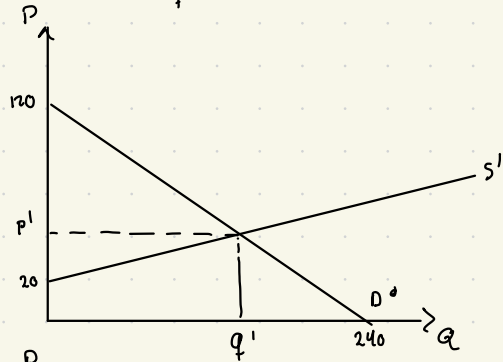
$$240 - 2p^0 = -80 + 4p^0 \Rightarrow q^0 = \frac{400}{3} \approx 133.33$$



$$2. \quad \text{Event 1 } Q_s^0 \rightarrow Q_s^1$$

$$Q_d^0(p^1) = Q_s^1(p^1) \Rightarrow p^1 = 40$$

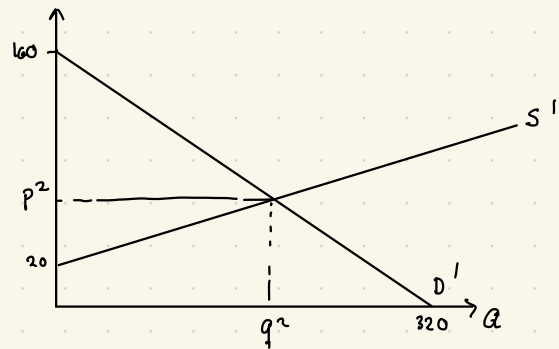
$$240 - 2p^1 = 8p^1 - 160 \Rightarrow q^1 = 160$$



$$3. \quad \text{Event 2 } Q_d^0 \rightarrow Q_d^1$$

$$Q_d^1(p^2) = Q_s^1(p^2) \Rightarrow p^2 = 48$$

$$320 - 2p^2 = 8p^2 - 160 \Rightarrow q^2 = 224$$

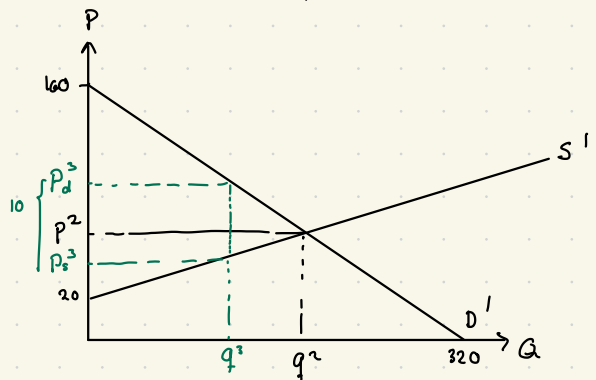


$$4. \quad \text{Event 3: } P_d^3 - P_s^3 = 10$$

$$Q_d^1(10 + P_s^3) = Q_s^1(P_s^3) \Rightarrow P_d^3 = 56$$

$$320 - 2(10 + P_s^3) = 8P_s^3 - 160 \Rightarrow P_s^3 = 46$$

$$q^3 = 208$$



$$5. \quad \text{Event 4: } Q_d^1 \rightarrow Q_d^2, \text{ and with tax } P_d^4 - P_s^4 = 10$$

$$Q_d^2(P_s^4 + 10) = Q_s^1(P_s^4) \Rightarrow P_d^4 = 64$$

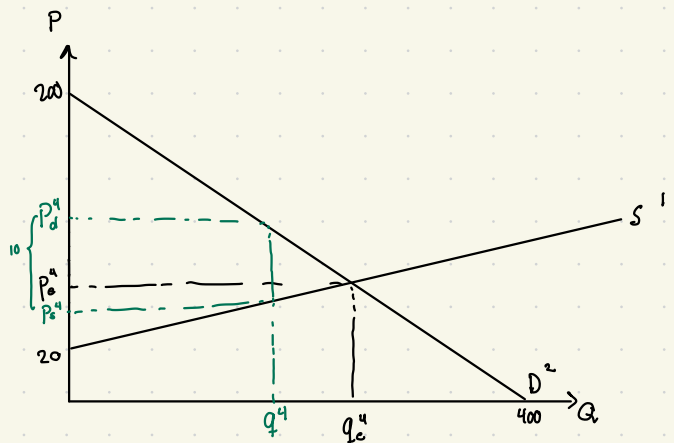
$$400 - 2(P_s^4 + 10) = 8P_s^4 - 160 \Rightarrow P_s^4 = 54$$

$$q^4 = 272$$

You will also need the eq. without tax: (q_e^4, p_e^4)

$$400 - 2p_e^4 = 8p_e^4 - 160 \Rightarrow p_e^4 = 56$$

$$q_e^4 = 288$$



6.

	Initial	Event 1	Event 2	Event 3	Event 4
CS	$\frac{40000}{9} \approx 4444.44$	6400	12544	10816	18496
PS	$\frac{2000}{9} \approx 222.22$	1600	3136	2704	4624
DWL	0	0	0	80	80

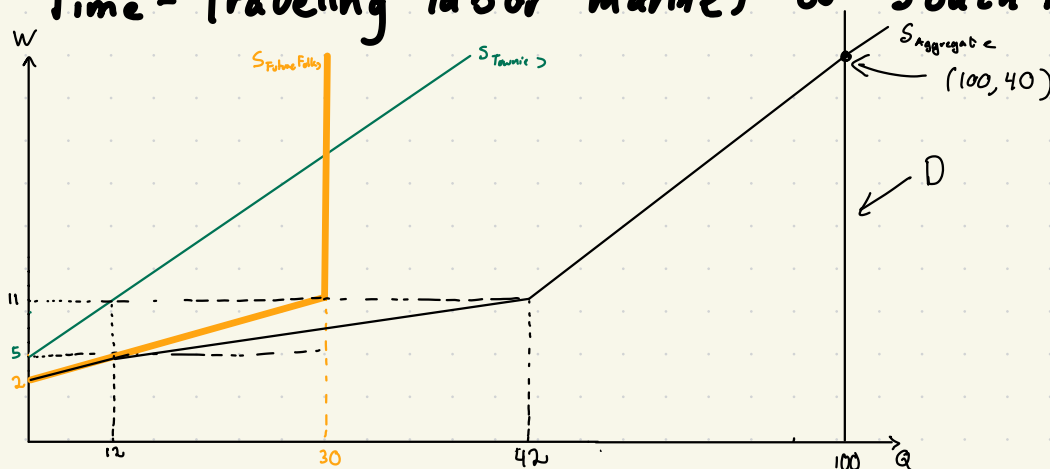
7.

Notice that in both Events 3 and 4, there is a deadweight loss since the tax generates a distortion in the equilibrium quantity: without the tax, a higher quantity of Sparkle Dust is consumed in both cases. Thus, there is a loss in efficiency with the tax compared to the unrestricted market equilibrium.

Furthermore, notice that the consumer incidence is $4/5$, while the producer incidence is $1/5$, i.e., consumers always pay 80% of the tax, while producers only pay 20%. Therefore, the change in welfare is always greater for the consumers than the producers in both events 3 and 4.

The Time-Traveling labor market of South Park

1.



$$S_{\text{Townies}}: W = 0.5Q_T + 5 \Rightarrow Q_T = \begin{cases} 2W - 10 & W \geq 5 \\ 0 & \text{otherwise} \end{cases}$$

$$S_{\text{FutureFolk}}: W = 0.3Q_F + 2, Q_F \leq 30 \Rightarrow Q_F = \begin{cases} \frac{10}{3}W - \frac{20}{3} & \text{if } 2 \leq W < 11 \\ 30 & \text{if } W \geq 11 \end{cases}$$

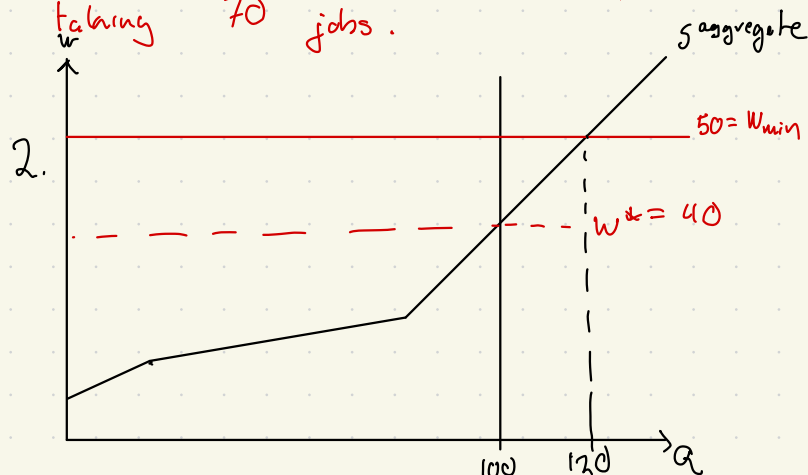
$$S_{\text{Aggregate}}: Q_T + Q_F = \begin{cases} \frac{10}{3}W - \frac{20}{3} & \text{if } 2 \leq W < 5 \\ \frac{16}{3}W - \frac{50}{3} & \text{if } 5 \leq W < 11 \\ 2W + 20 & \text{if } W \geq 11 \end{cases}$$

$$\Rightarrow \text{If } Q^* = 100 \Rightarrow W^*: 100 = 2W^* + 20 \Rightarrow W^* = 40$$

At $w^* = 40$ $Q_T^* = 70$
 $Q_F^* = 30$

Note that without future folk, Townies would offer 100 of jobs at a wage of 55.

Future Folk did take 30 jobs from townies; however townies are still taking 70 jobs.



At $w_{\min} = 50$ (it is an effective min wage) total supply is 120, in particular

$$Q_F(50) = 30, Q_T(50) = 90$$

(a) Method 1: Townies offer a proportion $\frac{90}{120} = 0.75$
 Future folk offer a proportion $\frac{30}{120} = 0.25$

$$\text{Thus with the minimum wage: } Q_T = 0.75 \times 100 = 75 \\ Q_F = 0.25 \times 100 = 25$$

(b) Method 2: Prioritizing townies, now we have that the first 40 jobs go to townies that want to work at a wage of 50, the remaining 10 go to future-folk. i.e., $Q_T = 90, Q_F = 10$

Note that the minimum wage does help townies with either method. Thus we conclude that after the policy townies should calm down 😊!